

Presentation Versus Ground Truth

A testable framework for virtual and real-world environments

Prepared for: Multi-AI collaborative pass-through

Date: September 3, 2026

Status: Conceptual research and design document; not a finding about any particular person, organization, or real-world incident.

Executive summary

The central problem is not whether to accept or reject an environment's presentation. It is how to distinguish **presentation**—what an agent, interface, institution, or sensor says is happening—from **recorded state and evidence**—what can be independently verified about events, actors, conditions, causes, and outcomes.

The most useful umbrella term for the proposed construct is **systemic presentation-layer deception**: an environment or its agents shape observable conditions, explanations, or viewpoints so that purposeful, coordinated, or causally related activity appears ordinary, unrelated, accidental, or emergent. This is a working term rather than a single universally standardized label. Its major mechanisms are environmental camouflage, signal dilution, causal masking, attribution obfuscation, narrative laundering, equivocation, and—where applicable—telemetry manipulation.

The corresponding audit method is **provenance-based causal reconstruction**. It does not assume that every unusual event is coordinated, and it does not treat an agent's explanation as irrelevant. Instead, it separates verified observations, derived causal relationships, inferred objectives, and predictions; then tests competing explanations against time-synchronized, independently retained evidence.

This framework is applicable to virtual worlds, simulations, games, distributed software, AI-agent testbeds, safety evaluations, workplace processes, customer-service settings, and other environments where the same event can be viewed through multiple partial representations. In real-world use, it must be an evidence and process framework—not a presumption that hidden intent or coordination exists.

1. Core question

The practical question is:

t; How can an observer determine whether an apparent environmental event is an independent occurrence, an emergent interaction, an ordinary action with an ordinary explanation, or an intentional action whose timing, context, or presentation has been shaped to conceal its role?

The question should be decomposed into four separate tests:

1. **Occurrence:** Did the event happen?
2. **Attribution:** Which actor, process, or condition produced or influenced it?
3. **Causation:** What earlier states or actions contributed to it?
4. **Intent:** Is there evidence that the event was selected or coordinated for a goal?

These should never be collapsed into one conclusion. A complete event record may establish occurrence and attribution without establishing motive. A repeated pattern may justify investigation without proving coordination.

2. The layered model

The concept is easiest to understand as five layers.

| Layer | Meaning | Testable object |

---|---|---

| Canonical state | The authoritative state maintained by the simulation or system | State variables, object positions, permissions, active processes |

| Event history | Ordered state transitions and actions that changed the state | Timestamps, event IDs, inputs, outputs, dependencies |

| Observation/rendering | What a participant, camera, interface, or sensor displayed | View-specific frames, audio, alerts, dashboards |

| Narrative/claim | What an agent or institution says the events mean | Statements, classifications, explanations, reports |

| Audit/reconciliation | Independent comparison of state, history, observations, and claims | Provenance graph, discrepancy log, alternative hypotheses |

A virtual environment can make the canonical state directly inspectable. A physical environment normally cannot provide a universal ground-truth ledger, so the practical equivalent is triangulation from independent records, witnesses, timestamps, policies, and artifacts.

The critical distinction is between an event and its interpretation. A door can be locked; a representative can say it is open; a camera can show the doorway; a transaction log can show when a lock command was issued. These are different evidentiary objects. NIST guidance describes audit records in terms of what happened, when and where it happened, the event source, outcome, and associated identities.[cite:10][cite:18]

3. Working terminology

Systemic presentation-layer deception

A condition in which agents, environmental processes, or system components shape the timing, composition, context, or apparent independence of observable events so that an underlying action, relationship, objective, or coordination pattern is interpreted as ordinary, unrelated, accidental, or emergent.

This term captures the whole construct while signaling that it is a proposed synthesis. It should not be presented as a settled name for a recognized legal or scientific diagnosis.

Environment-mediated causal masking

A subtype in which the environment itself carries the concealment. Individual events may be real and accurately recorded, but their relationship is obscured. For example, a noisy venue, traffic delay, crowd movement, or interface interruption may be used as the apparent explanation for a targeted timing or disruption pattern.

Attribution obfuscation

A strategy that makes it difficult to connect an event to the responsible actor, controller, process, or objective. In cybersecurity, the related concept of masquerading involves making an artifact appear legitimate or benign to users or defenses.[cite:19]

Signal dilution and noise injection

Signal dilution reduces the visibility of a relevant pattern by surrounding it with irrelevant or distracting activity. Noise injection adds or amplifies variability. These terms are strongest when the system can show that the extra activity was deliberately generated, scheduled, or selected; otherwise, “environmental variance” or “confounding activity” is safer.

Plausible deniability and narrative laundering

Plausible deniability preserves an innocent explanation for each individual act. Narrative laundering is a proposed descriptive term for translating an intentional or attributable action into an innocuous semantic category such as “traffic,” “restaurant noise,” “a glitch,” or “coincidence.” The event may remain visible while agency and causal meaning are obscured.

Equivocation and Byzantine behavior

Equivocation occurs when a component supplies inconsistent versions of events to different observers. Byzantine behavior is a broader distributed-systems category for arbitrary, inconsistent, or adversarial behavior by a node. These terms apply when the system can demonstrate inconsistent outputs or states; they should not be used merely because observers disagree.

Full-state observability and provenance

Full-state observability means access to internal state beyond a local rendering. Provenance records the entities, activities, and relationships involved in producing or influencing an event or artifact; the W3C PROV model is a formal reference point for representing and exchanging such information.[cite:4][cite:12]

Plan recognition and action prediction

A HUD that estimates what an agent may do next is better described as plan recognition, goal inference, policy inference, or next-action prediction—not mind reading. It must visibly distinguish predictions from verified historical actions. A probability of an action is evidence for investigation, not proof that the action occurred or that a particular motive existed.

4. The McDonald's example as a neutral test case

The restroom example illustrates a presentation-to-state mismatch without requiring hidden coordination. The representative said the restroom was open, but the observable operational condition was that the door remained locked until the representative activated and held the button. The mirror and the door provided a way to verify timing: the button needed to be activated while the customer was at the door, not merely asserted to be open.

A neutral reconstruction would be:

- **Claim:** “It is open; go over there.”
- **Observed condition:** The door did not open on the first attempt.
- **Operational dependency:** Access required activation of a button.
- **Verification action:** The representative remained at the control, observed the customer's position, held the button, and the door then opened.

• **Unresolved question:** Whether the initial statement resulted from misunderstanding, inattention, impatience, or deliberate misrepresentation.

The evidence establishes a claim–condition mismatch. It does not, by itself, establish motive. This is exactly why a good framework distinguishes **what was said**, **what happened**, **what condition controlled the outcome**, and **what remains unknown**.

5. Architecture for a virtual environment

A simulation designed to test this problem should expose both participant views and an authoritative audit plane.

```text

Canonical simulation state

|

+-- State-transition ledger

+-- Identity and privilege records

+-- Event provenance graph

+-- Environment-generator records

+-- Agent policy / action queue

|

+-- Participant rendering A

+-- Participant rendering B

+-- Sensor or camera rendering C

+-- Agent narratives and reports

|

Independent auditor

|

+-- Reconcile observations with canonical state

+-- Trace event causes and dependencies

+-- Compare claims with actions

+-- Label confidence and unresolved alternatives

...

The event model should record at least:

| Field | Purpose |

|---|---

| Event ID | Enables exact cross-reference across logs |

| Time | Supports ordering and temporal tests |

| Source | Identifies the process, agent, sensor, or subsystem |

| Location/context | Establishes where and under what conditions the event occurred |

| Input/trigger | Records what initiated the event |

| Action/state change | Describes what changed |

| Outcome | Records success, failure, or measurable effect |

| Authority | Shows which permissions or policies were used |

| Parent/dependency IDs | Connects the event to preceding causes |

| Integrity metadata | Detects later alteration or deletion |

| Confidence/provenance status | Separates verified, derived, inferred, and predicted content |

OpenTelemetry provides a practical software analogy: distributed traces consolidate events from multiple components, and traces can represent parent–child relationships across process, network, and security boundaries.[cite:3][cite:7][cite:11] The W3C PROV model

provides a general vocabulary for entities, activities, and agents involved in producing or influencing an item.[cite:4]

## 6. Testability framework

### Step 1: Define the hypothesis

Write the claim in a form that could be disconfirmed. Examples:

- “The interruption was generated by a named subsystem rather than by an independent environmental process.”
- “The agents' arrival times are more tightly coupled to the target task than to ordinary traffic conditions.”
- “The public rendering omits or misclassifies a state transition present in the canonical ledger.”
- “The access-control statement conflicts with the recorded lock state and control input.”

Avoid starting with an unfalsifiable claim such as “everything is coordinated.”

### Step 2: Define the unit of analysis

Choose an event window, target task, location, or transaction. Pre-register the start and end conditions. A unit might be one conference call, one simulated journey, one customer-service interaction, one audit, or one game episode.

### Step 3: Collect independent channels

Use at least two channels that are not controlled by the same actor or process. In software this may include append-only application logs, host telemetry, network traces, access-control records, and simulation-engine state. In real-world settings it may include receipts, timestamps, written policies, contemporaneous notes, independent witnesses, publicly available records, or the establishment's own documented procedure.

NIST recommends protecting audit information and tools from unauthorized modification or deletion, and reviewing records for unusual or inappropriate activity.[cite:18] NIST incident-response guidance treats collected incident data as evidence and emphasizes retaining it for analysis.[cite:26]

### Step 4: Synchronize and preserve

Use a common clock or document clock offsets. Preserve original files, hashes, chain-of-custody details where appropriate, and the distinction between raw evidence and analyst annotations. Never silently overwrite an earlier interpretation.

### Step 5: Reconstruct the event graph

Represent events as nodes and dependencies as directed edges. Use explicit links when available: shared transaction IDs, parent process IDs, causal triggers, control commands, or simulator event references. Treat temporal proximity alone as a weak relationship.

### Step 6: Compare competing explanations

At minimum, test:

1. Ordinary independent occurrence.
2. Emergent interaction without a shared objective.
3. Single-agent action with incidental side effects.
4. Coordinated action.
5. Telemetry or presentation error.
6. Incomplete or corrupted evidence.

For each explanation, list predictions, supporting observations, contradictions, and missing evidence.

### Step 7: Score evidence without converting scores into certainty

A practical evidence matrix can use qualitative labels:

| Evidence level | Meaning |

|---|---|

| E0: Claim only | An agent or observer reports an explanation |

| E1: Single observation | One channel records an event |

| E2: Independent corroboration | Two or more independent channels agree |

| E3: Attribution | An identity, process, or control path is linked to the event |

| E4: Causal linkage | A dependency or intervention connects action to outcome |

| E5: Coordination evidence | Shared control, communication, repeated coupling, or common task is demonstrated |

| E6: Intent evidence | Objective or selection rationale is directly documented or strongly established |

The level should be attached to each proposition separately. “Agent X caused event Y” may be E4 while “Agent X intended to obstruct task Z” remains E0–E2.

## 7. Metrics and experimental designs

### *Presentation–state discrepancy rate*

Measure how often participant-facing claims or renderings disagree with canonical state:

$$\lfloor$$

$$D_{\text{ps}} = \frac{\text{number of material presentation-state discrepancies}}{\text{number of audited events}}$$

$$\rfloor$$

A discrepancy must be defined in advance. A harmless display delay should not be treated as deception.

### *Attribution completeness*

Measure the proportion of material events for which the system can establish source, time, location, outcome, and associated identity. These fields align with NIST audit-record guidance.[cite:10][cite:18]

### *Causal trace coverage*

Measure the proportion of consequential events with an identifiable trigger or dependency path. Missing links are not proof of concealment; they are observability gaps.

### *Narrative–trace mismatch rate*

Compare stated explanations with recorded state and event provenance. Categorize mismatches as error, ambiguity, stale information, policy conflict, or suspected deception. Do not place all mismatches in the last category.

### *Coordination indicators*

Potential indicators include shared controllers, common task identifiers, unusual temporal coupling, repeated synchronized behavior, information exchange, coordinated response to interventions, or outcomes that consistently benefit a common objective. Each indicator needs a baseline and a comparison condition.

### *A/B and intervention tests*

In a simulator, randomly vary the environment while holding agent policies constant, then vary agent policies while holding environmental parameters constant. If an outcome follows the environment, it supports an environmental explanation; if it follows a hidden controller or policy change, it supports an agent/system explanation. Randomization and intervention are stronger than retrospective pattern matching.

### *Red-team/blue-team evaluation*

- **Red team:** Attempts to make purposeful activity appear ordinary without altering canonical logs.
- **Blue team:** Uses independent telemetry, provenance, and reconciliation to identify the activity.
- **Control group:** Runs ordinary activity with no adversarial objective.
- **Success criteria:** Detection rate, false-positive rate, time to attribution, causal-trace completeness, and resilience to viewpoint manipulation.

## 8. Real-world applicability and limits

In real organizations, the framework can improve incident review, quality assurance, workplace process analysis, customer-service escalation, safety investigations, algorithmic auditing, and governance. It is particularly useful where a frontline explanation, dashboard, or public-facing interface may not expose the operational condition that produced an outcome.

The real-world version should not claim access to private thoughts or universal ground truth. It should rely on observable records, authorized system access, lawful data collection, independent corroboration, and a documented opportunity for correction. NIST’s AI Risk Management Framework is designed to help organizations manage risks and incorporate trustworthiness considerations across the AI lifecycle.[cite:5]

The framework should also preserve privacy and proportionality. Full-state observability in a simulation does not imply that a real employer, business, or individual may collect every possible data point. Data minimization, access controls, retention limits, purpose limitation, and independent review are necessary design constraints.

## 9. Failure modes

### ***Over-correlation***

Several events occurring close together do not establish coordination. Traffic, crowding, interruptions, and delays can be genuinely independent or emergent. Require causal links, repeated controlled patterns, shared control, or intervention evidence before escalating.

### ***Prediction treated as history***

A high-probability next-action prediction is not a record that the action occurred. The HUD should use distinct visual labels for verified, derived, inferred, and predicted information.

### ***Sensor monoculture***

Multiple cameras or logs controlled by one source may create the appearance of corroboration while sharing one failure or manipulation point. Independence must be evaluated, not assumed.

### ***Log completeness mistaken for truth***

Logs can omit events, misattribute actions, suffer clock drift, or be altered. Audit integrity and coverage must be measured. A canonical ledger is only as trustworthy as the simulation's implementation and its protection from unauthorized change.

### ***Motive inflation***

An operational mismatch may reflect confusion, training failure, bad interface design, disability access issues, or ordinary error. Intent should be the last inference, not the first.

### ***Confirmation bias in the auditor***

A privileged observer can create false narratives by selecting only confirming events. The audit process should require alternative hypotheses, disconfirming evidence, blind review where feasible, and explicit uncertainty.

## **10. Operational template**

Use the following template for each incident, test, or simulation episode.

### ***Event record***

- **Case ID:**
- **Environment/system:**
- **Task or objective under review:**
- **Start/end time:**
- **Observed presentation:**
- **Agent or institution explanation:**
- **Concrete state or outcome:**
- **Independent records available:**
- **Known identities/processes:**
- **Relevant permissions or controls:**
- **Candidate triggers:**
- **Observed dependencies:**
- **Competing explanations:**
- **Evidence level for occurrence:**
- **Evidence level for attribution:**
- **Evidence level for causation:**
- **Evidence level for intent:**
- **What would disconfirm the leading explanation:**
- **Unresolved questions:**
- **Corrective action:**
- **Review date and reviewer:**

### ***Decision rule***

Do not ask only, “Does the presentation sound plausible?” Ask:

1. What was presented?
2. What was independently recorded?
3. What state transition occurred?
4. Who or what could produce it?
5. What causal links are demonstrated?
6. Which explanation best fits all channels?
7. What alternative explanation remains viable?
8. What intervention or additional record would distinguish them?
9. What corrective action addresses the underlying condition regardless of motive?

## 11. Recommended final terminology

For collaborative use, the following vocabulary is recommended:

| Function | Preferred term |

|---|---|

| Whole phenomenon | Systemic presentation-layer deception |

| Environmental mechanism | Environment-mediated causal masking |

| Hidden relationship | Causal masking |

| Hidden responsibility | Attribution obfuscation |

| Added distraction | Signal dilution or deliberate noise injection |

| Benign-seeming activity | Environmental camouflage |

| Conflicting accounts | Equivocation |

| Adversarial node behavior | Byzantine behavior |

| Reliable history | Canonical state-transition record |

| Origin and dependencies | Event provenance |

| Cross-system reconstruction | Distributed tracing and event correlation |

| Full internal access | Privileged full-state observability |

| Likely goal | Plan recognition or goal inference |

| Likely next action | Policy or next-action prediction |

| Audit method | Provenance-based causal reconstruction |

The single-sentence definition is:

***t; Systemic presentation-layer deception is the shaping of observable environmental conditions and explanations so that purposeful or causally related activity appears ordinary or accidental, while provenance-based causal reconstruction tests that presentation against independently preserved state transitions, identities, dependencies, and outcomes.***

## 12. Research basis and scope

The framework is synthesized from established practices rather than presented as one pre-existing named phenomenon. NIST security guidance defines the content and protection expectations for audit records; OpenTelemetry operationalizes distributed traces and metrics; W3C PROV formalizes provenance; MITRE ATT&CK documents masquerading as a defense-evasion technique; and NIST’s AI RMF provides a governance context for measuring and managing AI-system risks.[cite:2][cite:3][cite:4][cite:5][cite:19]

The proposed umbrella terms—particularly “systemic presentation-layer deception” and “provenance-based adversarial observability”—are design vocabulary for this collaborative framework. They should be treated as precise working labels, not as claims that a standards body or research field has adopted those exact phrases.